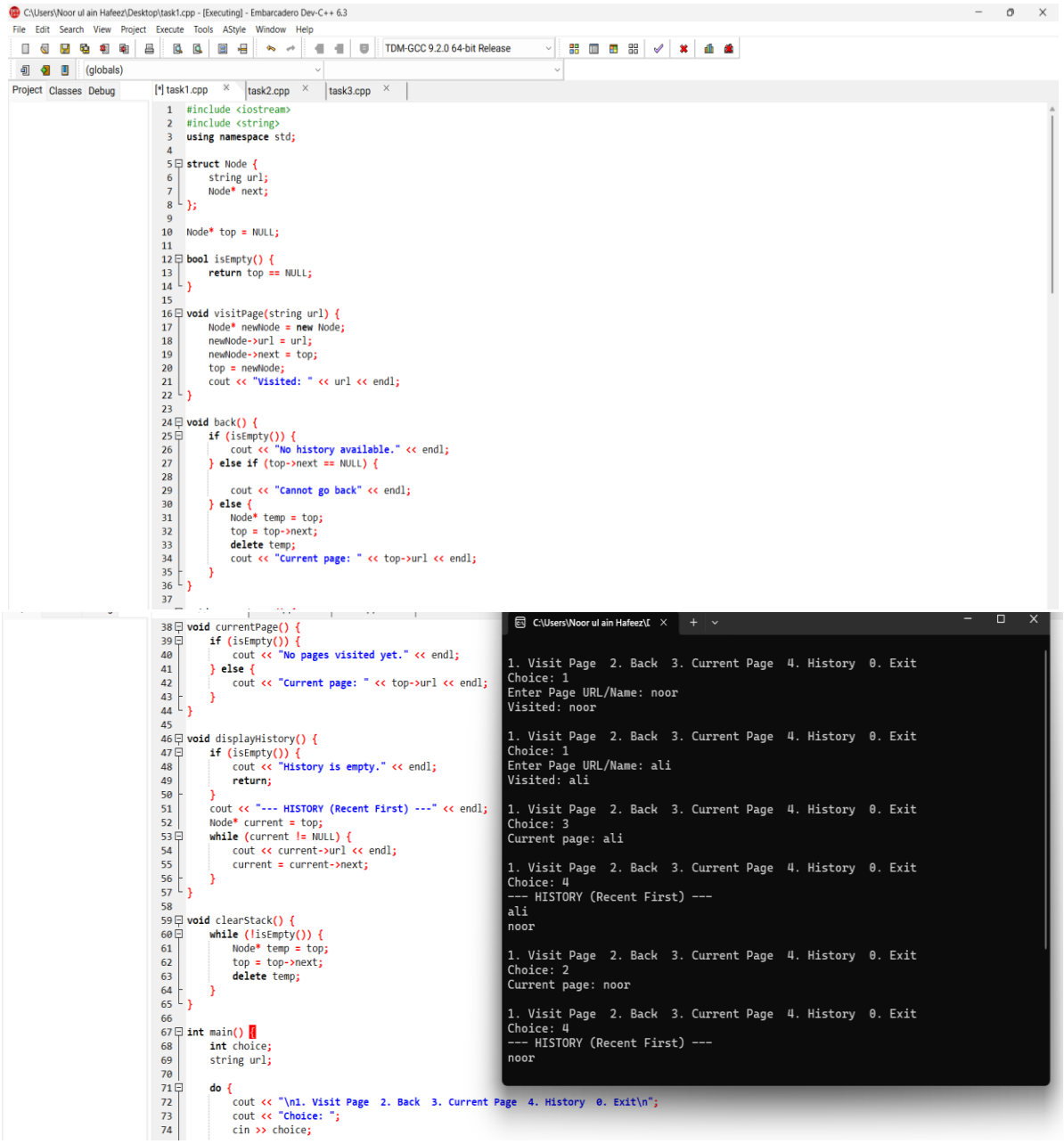
DSA Lab: 10

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Task 1:



```cpp
#include <iostream>
#include <string>
using namespace std;

struct Node {
    string url;
    Node* next;
};

Node* top = NULL;

bool isEmpty() {
    return top == NULL;
}

void visitPage(string url) {
    Node* newNode = new Node;
    newNode->url = url;
    newNode->next = top;
    top = newNode;
    cout << "Visited: " << url << endl;
}

void back() {
    if (isEmpty()) {
        cout << "No history available." << endl;
    } else if (top->next == NULL) {

        cout << "Cannot go back" << endl;
    } else {
        Node* temp = top;
        top = top->next;
        delete temp;
        cout << "Current page: " << top->url << endl;
    }
}

void currentPage() {
    if (isEmpty()) {
        cout << "No pages visited yet." << endl;
    } else {
        cout << "Current page: " << top->url << endl;
    }
}

void displayHistory() {
    if (isEmpty()) {
        cout << "History is empty." << endl;
        return;
    }
    cout << "--- HISTORY (Recent First) ---" << endl;
    Node* current = top;
    while (current != NULL) {
        cout << current->url << endl;
        current = current->next;
    }
}

void clearStack() {
    while (!isEmpty()) {
        Node* temp = top;
        top = top->next;
        delete temp;
    }
}

int main() {
    int choice;
    string url;

    do {
        cout << "\n1. Visit Page  2. Back  3. Current Page  4. History  0. Exit\n";
        cout << "Choice: ";
        cin >> choice;
```

```
1. Visit Page  2. Back  3. Current Page  4. History  0. Exit
Choice: 1
Enter Page URL/Name: noor
Visited: noor

1. Visit Page  2. Back  3. Current Page  4. History  0. Exit
Choice: 1
Enter Page URL/Name: ali
Visited: ali

1. Visit Page  2. Back  3. Current Page  4. History  0. Exit
Choice: 3
Current page: ali

1. Visit Page  2. Back  3. Current Page  4. History  0. Exit
Choice: 4
--- HISTORY (Recent First) ---
ali
noor

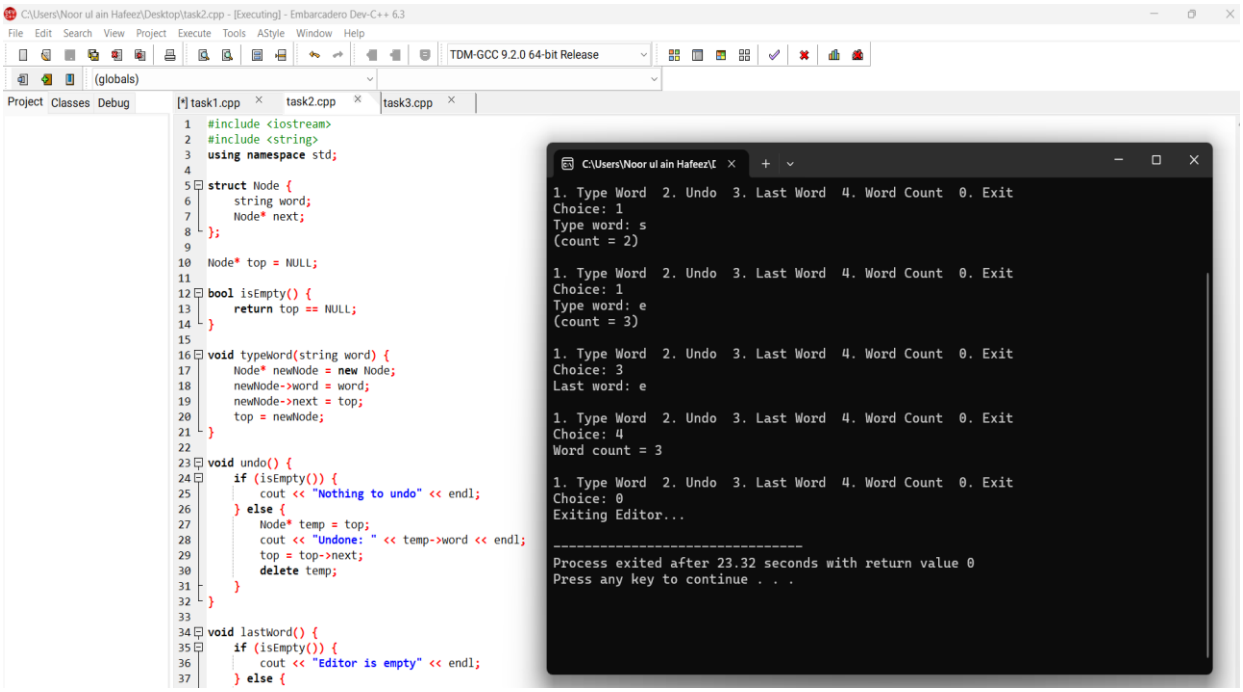
1. Visit Page  2. Back  3. Current Page  4. History  0. Exit
Choice: 2
Current page: noor

1. Visit Page  2. Back  3. Current Page  4. History  0. Exit
Choice: 4
--- HISTORY (Recent First) ---
noor
```

```
        switch (choice) {
            case 1:
                cout << "Enter Page URL/Name: ";
                cin >> url;
                visitPage(url);
                break;
            case 2:
                back();
                break;
            case 3:
                currentPage();
                break;
            case 4:
                displayHistory();
                break;
            case 0:
                clearStack();
                cout << "Exiting Browser..." << endl;
                break;
            default:
                cout << "Invalid choice!" << endl;
        }
    } while (choice != 0);

    return 0;
}
```

Task 2:



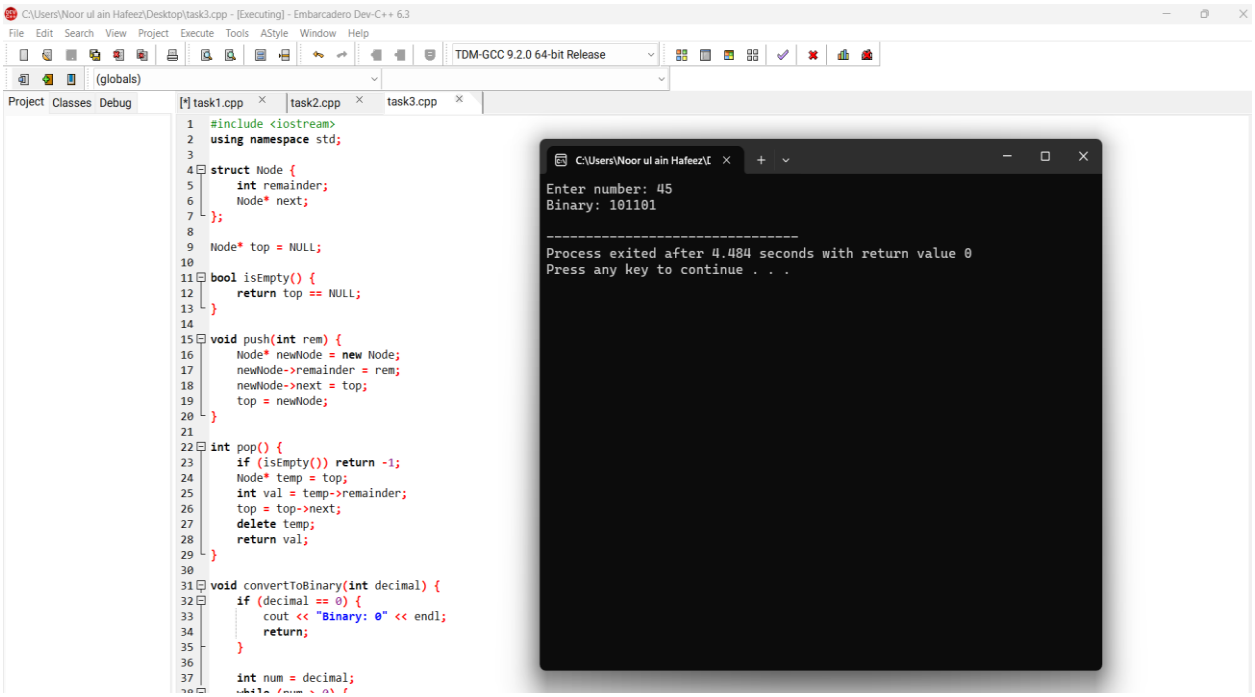
```cpp
int main() {
    int choice;
    string word;

    do {
        cout << "\n1. Type Word  2. Undo  3. Last Word  4. Word Count  0. Exit\n";
        cout << "Choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                cout << "Type word: ";
                cin >> word;
                typeWord(word);
                cout << "(count = " << wordCount() << ")" << endl;
                break;
            case 2:
                undo();
                cout << "(count = " << wordCount() << ")" << endl;
                break;
            case 3:
                lastWord();
                break;
            case 4:
                cout << "Word count = " << wordCount() << endl;
                break;
            case 0:
                clearStack();
                cout << "Exiting Editor..." << endl;
                break;
            default:
                cout << "Invalid choice!" << endl;
        }
    } while (choice != 0);

    return 0;
}
```

Task 3:



```
    while (num > 0) {
        push(num % 2);
        num /= 2;
    }

    cout << "Binary: ";
    while (!isEmpty()) {
        cout << pop();
    }
    cout << endl;
}

int main() {
    int num;
    cout << "Enter number: ";
    cin >> num;

    if (num < 0) {
        cout << "Please enter a non-negative integer." << endl;
    } else {
        convertToBinary(num);
    }

    return 0;
}
```